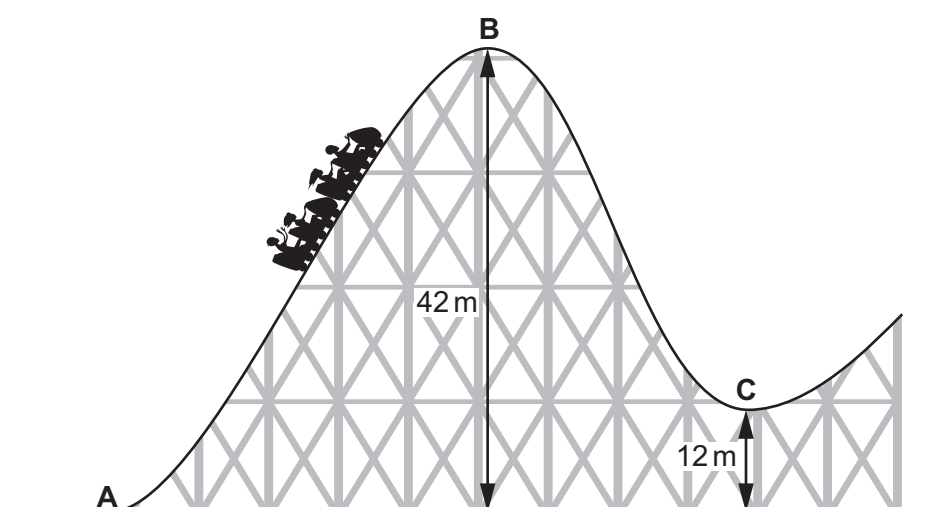


7. The diagram shows part of a rollercoaster ride at a theme park.



- (a) A motor with a power output of 65 kW and a chain mechanism pulls the carriages of mass 2 600 kg from **A** to **B** in a time of 32 s.

(i) Show that the work done by the motor in 32 seconds is approximately 2 MJ. [1]

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(ii) Hence calculate the efficiency of the mechanism, assuming the carriages are momentarily at rest at **B**. [3]

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- (b) At B, the carriages become disconnected from the motor and the carriages move under the influence of gravity for the rest of the ride. In moving from B to C, a distance along the track of 36 m, the carriages experience a mean resistive force of 2.8 kN. Calculate the speed of the carriages at C. [5]

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END OF PAPER

